SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Thamaraiselvi.S (192511042) (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Thamaraiselvi

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage %	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

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B.E CSE

CSA1708 - Artificial Intelligence.

(CO1) Assessment Tool - 3

1) Smart Home Automation system:

Types of Intelligent Agents

a) simple Reflex Agent:

* works using condition-action rules.
* Makes decisions based only on the current input.
* example: if motion is detected -> Turn on lights.

Advantages:

* fast response.
* Easy to implement.

Disadvantages:

* Cannot remember previous events.
* Cannot learn user preferences.

b) Model-Based Agent:

* Maintains an internal model of the environment
* uses past and current information for decisions.

* example: Remembers that no one is at home and switches of unecessary lights

Advantages:

* Handles partially observable environments.
* Better decision-making.

Disadvantages:

* Requires more memory and computation.

c) <u>Goal-Based Agent</u>:

* Acts to Acheive a specific Goal.
* Evaluates different actions before choosing one.
* example: Keeps the home temperature at 24°c while minimizing electricity usage.

Advantages:

* flexible and intelligent.
* Chooses the best path to acheive goals.

Disadvantages:

* slower than reflex agents.

* Responds in real time.
* Saves electricity.
* Maximize comfort.

Conclusions:

A hybrid intelligent agent offers the best balance of comfort, energy efficiency and security for a smart home automation system.

2) Robot in an Unknown Maze:

Uniform Cost search (UCS):

Definition:

UCS expands the node with the lowest cumulative path cost.

Advantages:

* Complete.
* Always finds the lowest cost path
* suitable when path costs are different.

Disadvantages:

* High memory usage.

d) Utility - Based Agent:

* Chooses the action with the highest utility (best overall benefit).

* Balances multiple objectives like comfort, security, and energy efficiency.

* Example: Adjust AC, lights and security setting to maximize comfort while reducing electricity bills.

Advantages:

* Produces optimal decisions.

* Handles conflicting objectives effectively.

Disadvantages:

* Utility function design is complex.

Justification:

A Hybrid Agent (Model - Based + Utility - Based) is the most effective.

Reasons:

* Learns for user habits.

Time

Sp

Time complexity : $O(b^{1+\lfloor C^*/\varepsilon \rfloor})$

Space complexity : $O(b^{1+\lfloor C^*/\varepsilon \rfloor})$

Iterative Deepening search (IDS)

Definition :

IDS repeatedly performs depth-limited search with increasing depth limits untill the goal is foun

Advantages :

* Complete.
* Requires very little memory.
* Finds the shallowest solution.

Disadvantages :

* Repeats searches at previous depths
* Not suitable when path costs vary.

Time complexity : $O(b^d)$

Space complexity : $O(bd)$.

Comparison :

Feature	UCS	IDS
Complete	Yes	yes
Optimal	yes (lowest)	Yes (equal)

High	low
Slower	faster for shallow goals
Different path cost	Equal path costs.

Relation to Intelligent Agents:

Simple Reflex Agent:

* Reacts only to current perception.
* Not suitable for unknown mazes.

Model-Based Agent:

* Builds a map while exploring.
* Remembers visited locations.

Goal-Based Agent:

* plans actions to reach the maze exit.

Utility-Based Agent:

* chooses paths based on cost, safety, and time.

Best combination:

* Recommended Agent: Goal Based + Model Based Agent.
* Recommended search Algorithm: uniform Cost search (UCS).